

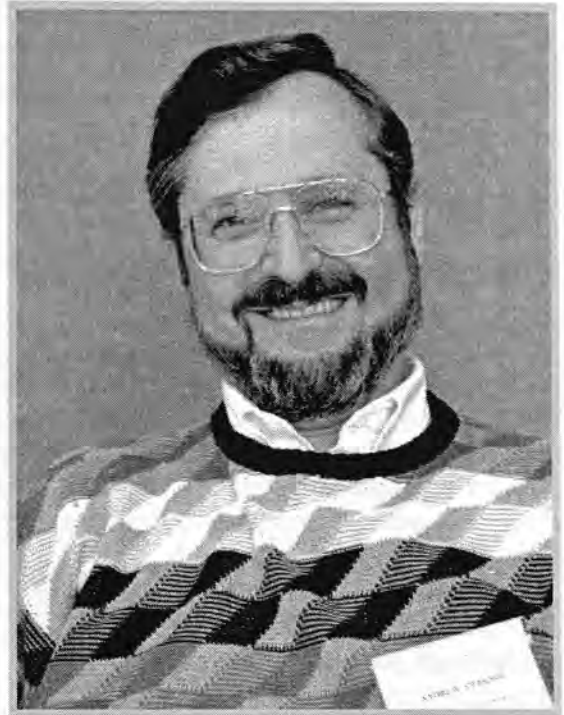
SETI, Education, News, and Entertainment

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Introduction

Detection of an ETI civilization by the NASA HRMS endeavor is likely to have diverse (and perhaps dramatic) effects on institutions and processes that inform and educate the public. Preparations for the full range of these effects and assessment of their likely magnitudes would entail activities and detailed studies of the kinds recommended at the end of this chapter. For present purposes, we can begin to anticipate some of the likely effects and responses by listing the kinds of institutions and processes that would probably be affected and by identifying historical examples of partially-analogous ideas, discoveries, and research whose past effects on educational or news-propagating processes can be documented. (Our analyses and recommendations apply almost exclusively to systems and events in the United States. Further studies of these issues as they apply in other cultural settings are essential, given that SETI research is international and that the findings are of relevance to people worldwide.)

In the following, we consider institutions that inform the public in three separate ways: education, dissemination of news, and entertainment. We focus initially on the roles of these institutions, then examine ways in which an ETI detection might influence their activities (and ultimately the public).



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Educational Institutions and Processes

Institutions and processes that inform and educate the western public can be grouped in three categories (Table 5-1). First are institutions of the formal educational network, the schools, colleges and universities whose primary missions are to convey knowledge.

These provide a mandatory agreed-upon foundation of education for young people and additional education for older students who seek it. The formal education network has an important research and scholarship component that discovers and creates new knowledge. The first availability of this new knowledge to society is often through student and faculty members of the university community.

Second are institutions that convey information to self-motivated persons who seek knowledge. These include museums, public lecture programs, membership organizations, public libraries, and other informal conduits of education.

A third group of relevant institutions includes society's educational infrastructure. Educators' and administrators' unions and associations, the textbook publishing industry, educational equipment distributors, government education departments and boards, and others that regulate and influence educational processes make up this component.

A comprehensive assessment of the effects of SETI research on public awareness, education, and informing would require examination of likely responses of these institutions.

Formal Educational Institutions

We define formal educational institutions as those that hold classes for students. The way that SETI educational programs would be of use to these institutions and the ways in which detection of an ETI signal might affect them would vary according to the age groups being served. Consequently we discuss the various levels separately.

Elementary and Secondary Schools

Although the fastest response to an announcement of an ETI detection would be made by the media, many schools and their support institutions would move swiftly to bring information about the discovery into the classroom. One important factor in speeding the discovery to classrooms is teacher interest. Teachers, like most people, are fascinated by the prospect of intelligent extraterrestrial life and recognize that their young students find few other topics as intriguing. Many classes would post news of an ETI detection on bulletin boards or as displays as soon as news of a verified detection was announced.

After the announcement of a verified ETI signal, science organizations, museums, professional societies, space agencies and others would issue materials with details of the discovery and perhaps some commentary about "what does it mean to science and humanity." Star maps would be readily available with the location and details of the ETI source. Slide sets and materials on astronomy, evolution, and other relevant topics would be updated immediately to deal with the discovery. Some of these materials would quickly enter classrooms. Adoption of textbooks is a slow process, however, and many years could elapse before revised textbooks reflecting the discovery would become generally available in classrooms.

Only one other subject compares with extraterrestrial life in its inherent fascination for teachers and young students alike. That topic – dinosaurs – is intensively studied at elementary school levels. Teachers conducting classes on dinosaurs have a wealth of information from which to select. By contrast, teachers in the ETI detection situation would almost certainly be working with limited information

Formal Institutions, Schools, Colleges, and Associated Programs	Informal Educational Institutions and Activities	Educational Infrastructure Institutions
Public Primary and Secondary Schools	Museums – Art, Historical, Science and Industry	Education Associations
Public Colleges and Universities	Outreach Programs (mainly from educational institutions, but including NASA)	Textbook Publishers
Public Graduate and Post-doctoral Institutions	Planetaria	Teacher Training Organizations
Private Schools, Colleges, Graduate Facilities	Public Libraries	Organizations that Drive Curriculum Reform
Home-study Programs (via TV, radio, correspondence)	Zoos, Aquaria, Botanical Gardens	Businesses that Manufacture and Distribute Slides, Science Equipment
	Membership Organizations and Clubs	Government Education Departments and Boards
	Scholarship and Research Facilities – “Think Tanks”	Educators and Administrators Unions
	Speakers Bureaus	Governing Bodies – Legislatures
		Private Foundations, Philanthropy
		Alumni Associations

Table 5-1. Institutions and Activities that Educate the Public

about any detected signal and would be forced to resort to scanty pre-existing material. Unlike paleontology, SETI science is paralleled by an extensive pseudoscientific “UFO industry” that generates volumes of unreliable material related to ETI’s. Teachers hastily seeking sources with which to inform their classes in the event of a discovery would

encounter this literature as well as more factual material and might not always be able to distinguish the reliable from the dubious. Here may be an opportunity for NASA to prepare for a flood of requests for reliable background information for “SETI bulletin boards” in classrooms and for presentations by teachers in schools nationwide.

A growing number of the nation's school teachers are members of professional associations such as the National Science Teacher's Association (NSTA). These active participants are exposed to new developments in the sciences about as rapidly as they occur. The NSTA and comparable bodies provide an important conduit that could be first on the scene with reliable detailed information about an ETI detection for teachers who maintain these contacts.

A few instances of SETI-related education in elementary school classrooms are known to us. For example, the widely-circulated *Weekly Reader*, a news bulletin for elementary school students, has featured the search for life in the universe in some issues (e.g., February 21 and April 10, 1992). A project funded by the National Science Foundation and NASA (and conducted by the SETI Institute) is currently devoted to developing science lessons based on the theme *Life In The Universe* for students in grades 3-9. The lessons teach standard science concepts in contexts that enlist the intense curiosity of young students about extraterrestrial life. The Drake Equation serves as the underlying multi-disciplinary theme throughout these lessons. A number of other materials are currently available for higher-level classes, for teachers who wish to bring ETI into their classrooms. These include:

- a) Curriculum materials from the Astronomical Society of the Pacific (ASP);
- b) A slide and information set on SETI assembled by Frank Drake and Andrew Fraknoi (20 slides with a 20-page booklet), now available through the ASP catalog and in use in over 1,500 schools and universities;
- c) The video *Quest for Contact* (which comes

with a resource guide to SETI for teachers) distributed by the SETI Institute, ASP, NASA, and other organizations;

- d) SETI updates in the Planetary Society's journal *The Planetary Report* (with circulation to 80,000 members);
- e) A bibliography of SETI-related information distributed by the Astronomical Society of the Pacific.

Most of these materials are readily adaptable to high school science classes and to certain classes in history and the social sciences.

Teachers are occasionally informed of SETI-related topics by regular periodicals. For example, the Spring, 1992 issue of the ASP's *The Universe in the Classroom* newsletter on teaching astronomy in grades 3-12 was entirely devoted to SETI and its use as a teaching tool. This issue went to 20,000 teachers in the US and is being translated into 6 languages other than English. Thus some materials and conduits of information are already available to secondary schools and teachers.

Colleges and Universities

Courses offered at colleges and universities are important conduits of information about SETI to the public. About 400,000 undergraduate students take astronomy courses for non-science majors each year. Most of the astronomy textbooks feature a section on SETI and/or exobiology. Such texts are revised frequently because the market for them is so large. A few schools have specialized courses or seminars on the subject of extraterrestrial life. Some of these courses are offered by one department, others are interdisciplinary in nature.

In addition to offering classes, colleges sponsor lecturers from professional societies that have SETI interests (for example, the American Astronomical Society and the American Institute for Aeronautics and Astronautics). These individuals give campus-wide talks on SETI, as do SETI scientists. The American Astronomical Society provides speakers for some 100 colleges each year, with concentration on those that lack astronomy departments. These individuals include SETI among their topics. Their lectures provide an important means for articulate and/or interested members of the public to learn about SETI research, the views of the researchers, and the cosmos. In another example, San Francisco State University, with funding provided by a grant from the California State University System, hosted a 10-week lecture series devoted to the Drake Equation and SETI science. Thus the institutions of higher learning already have several mechanisms in place for transmitting the most recent information available about any newly detected ETI signal to sizable numbers of receptive people.

Informal Educational Institutions and the Educational Infrastructure

Many of the organizations in the second and third columns of Table 5-1 play a much more important role in education (and particularly in science education) than is generally appreciated. In addition, membership organizations such as the Astronomical Society of the Pacific and the Planetary Society provide reliable lines of communication to many members. Many of these institutions respond rapidly to new developments and discoveries with attractive displays and information services. Because libraries, planetaria and museums are destinations for people

with children, they serve a large and important sector of the motivated public. Informal educational institutions are the source of first resort for many people who seek updated information on recent scientific developments. In the following, we discuss a few examples of these institutions.

Museums and Planetaria

No reviews or surveys of SETI-related programs or displays at museums and planetaria have been compiled. However, there have been a substantial number of such programs over the years. Possibly the most prominent among the planetarium shows, at least in terms of number of attendees, is the *Calling All Stars* program that ran for two years (1989-91) at the National Air and Space Museum. SETI researchers have assisted with the development of planetarium shows at St. Louis, San Diego, and elsewhere. An example of an impressive SETI-related museum exhibit is the "Where Next, Columbus?" show at the National Air and Space Museum, scheduled to run for two years from December, 1992.

Programs offered by museums and planetaria tend to be exceptionally attractive and even awe inspiring. For example, the IMAX (large-screen) film *Blue Planet* gives viewers a sense of the Earth that is unobtainable in any other way. *Blue Planet* has been seen by ten million viewers, worldwide. One could argue persuasively that a comparable large-screen film about humanity's place in the cosmos and the possible existence of intelligent life elsewhere could be the most positive short-term popular effort that NASA could undertake. Screened in advance of any detection, such a film would prepare the way for public understanding in depth of the ramifications of the

detection. Breathtaking and attractive to viewers in many ways, such productions compete powerfully with science fiction films and have the advantage of portraying only those technologies that are realistically attainable. Other supremely attractive educational display possibilities undoubtedly await identification as well.

Public Libraries

As a public educational resource, good libraries are in a class by themselves. They provide a repository of easily-accessible written information that is increasingly replicated in video, audio, Braille, and other alternative modes. The custodians are motivated to separate factual from fictional material and to maintain high standards. Libraries are more widely dispersed and numerous than museums and more accessible to children unaccompanied by parents. They make routine use of network services to locate materials for individuals who request them. Many libraries also offer after-school reading programs for youngsters.

No reviews or surveys of existing SETI-related programs or displays at libraries are known to us. Our experience provides anecdotal evidence of interest, however. The SETI Institute has aided in two library displays in the past, both of which used visual aids (posters and artwork) that were loaned to the libraries by the Institute. Both libraries expressed appreciation and said they considered the displays a success. Both libraries already had many high quality SETI-related books on their shelves and used these as part of their displays. In general, the limited displays offered by libraries are seldom as grandly staged as those of museums.

In these ways, libraries are a reliable source of

accurate information that is easily accessible to almost all Americans. On the downside, their educational influence may be dwindling as a result of a general decline in the reading habits of the public.

A useful data set for persons interested in the impact of an ETI detection event (and public interest in SETI) would be a survey of SETI-related books currently on library shelves. This might include a comparison of the numbers of such books with those pertaining to UFO's and similar phenomena, and (if records are accessible) surveys of the numbers of people who check out both kinds of volumes. These books will be on the front line if a stampede for SETI-related information is unleashed by an ETI detection.

A judicious preparation by NASA for a sudden increase in inquiries might well include identification (or publication) of a volume intended for library shelves, specifically intended to be useful in the event of an ETI detection. (This volume could serve the additional purpose of being the source most highly recommended for teachers.) Communication between NASA and librarians would be easy via their professional newsletters, and would ensure that they are aware of updated SETI-related bibliographies and videotapes.

Effects of an ETI Detection on Educational Institutions

Changes in Elementary and Secondary School Education

Perhaps the most important effect of an ETI discovery would be a slow-developing but pervasive one, acting on the public through educational institutions via the steady growth of the population of persons who have heard



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about the discovery from the time they started school. For them, the existence of ETI's would assume equal stature with other topics and concepts routinely encountered there – for example, atoms, dinosaurs, and the faraway places encountered in geography lessons. As experience with teaching evolution demonstrates, this does not necessarily lead to widespread acceptance or understanding. For some children, however, the fact that a topic is part of a school curriculum gives it a separate legitimacy that ideas presented in newspapers or entertainment media lack.

Knowledge of ETI existence acquired in school would play an important part in views of life developed later. Not the least important of the practical outcomes would be a realization that humanity must think in terms of future generations, rather than immediate grat-

ification, in planning and conducting any two-way communication with the ETI's.

Another important effect may result from a way in which SETI-based teaching can assist science education. SETI is an exceptionally interdisciplinary field of research. Its scientists draw upon cosmology, astronomy, planetary and Earth sciences, biology, physics, chemistry, meteorology and a host of other subjects in formulating their views. The interdisciplinary nature of SETI has both advantages and disadvantages for teaching children about science.

On the one hand, a SETI-based approach provides a perfect opportunity to teach science in an integrated way. The above disciplines, with math, engineering, and psychology, can be blended in a SETI-oriented curriculum with important facts about history, philosophy, religion and other disciplines from the humanities. SETI is also an ideal forum to explore issues about doubt, uncertainty, and how science does not always mean the attainment of absolute truth. These positive features are recognized in the California State Framework for science teaching, which includes the following language;

Is there life elsewhere in the universe? At present any discussion must be highly speculative. Neither living organisms nor the traces of extinct living organisms have been found to date on any of the planets and moons we have visited with space probes. However, one can argue plausibly that life is possible elsewhere. Although no planets have been definitely detected around other stars, there are vast numbers of sunlike stars in the universe. Among the many planets they may have, there may be many having conditions much like those on earth. On at least some of those, some kind of life may have evolved; the laws of nature are the same everywhere in the uni-

verse. At present, however, we know very little. If life exists outside our solar system, we could probably not detect it in the foreseeable future unless it included intelligent beings who transmitted electromagnetic signals. There has been much discussion about whether we could detect such signals. Searches for them by NASA have been unsuccessful to date, but more extensive efforts can be expected in the future. In any case we cannot at present expect to have direct contact with any intelligent beings that may exist because of the vast distances involved. (California Department of Education, 1990, p. 86).

As a practical matter, curricula and testing as done currently in most schools are not very amenable to interdisciplinary innovations. The curriculum is already crowded with material, and teachers are not trained to teach content from a wide variety of disciplines in an integrated way. In addition, secondary school textbooks are currently very "single-subject" in content and most school districts cannot afford to buy books very often. It is also possible that certain groups may oppose the adoption of SETI-related curricula for various reasons. Parents, school boards, or teachers may be concerned that interdisciplinary curricula will lower student scores on standardized national tests. Special interest groups (such as some religious organizations) may oppose the notion of spending time in the classroom on extraterrestrial life.

To summarize, the fact of discovery of an ETI civilization would eventually become internalized by the first generation of children to learn about it in school, with probable (but perhaps unpredictable) shifts in their views of life. Any resultant increase in ETI-associated teaching could enhance education in other ways, as noted.

The Reaction of University Communities

A signal from another world would affect university communities in an important and predictable way. That is, it would trigger a wave of research and scholarship. This would in turn affect public perceptions of SETI in various ways.

Although SETI researchers have their own views of criteria for confirming that an artificial-appearing signal is genuinely of ETI origin, scholars and researchers would certainly rush to disprove it. Some scientists would invest considerable effort in trying to show that a simple "dial tone" was the product of some natural process occurring out in the universe. These persons and their arguments would be prominent in the public discussion following detection. Even a seemingly-indisputable, complex, information-rich message – "GREETINGS, EARTHLINGS" – or equivalent – would initially (and properly) be suspect as a hoax.

Initial scientific disagreement on the question of genuine ETI origin of the signal would almost certainly be confusing and irritating to a public that (mistakenly) expects unambiguous pronouncements from science. This post-detection skepticism could, however, also contribute positively to public information and education by presenting interested persons with descriptions of star activity and other natural events in space that would need to be evaluated by anyone seeking to draw their own conclusions about the origin of the "dial tone." Depending upon the circumstances, a well argued counterproposal could be exceptionally instructive to the scientific community and the public at large, both by enlarging public familiarity with phenomena in space and by highlighting the nature of scientific dialog.

The research of scholars may swing scientific opinion toward the view that a detected (genuine) ETI signal is truly of artificial origin. On the other hand, the research may succeed at suggesting enough plausible alternative hypotheses that the issue remains clouded for many years or decades. In the latter case, public doubt and uncertainty will persist for a very long time. If it is possible to decisively disprove the alternatives, leaving intact the hypothesis that the signal is truly of artificial origin, this long scientific catharsis will eventually pay off in deeply rooted public confidence that the existence of extraterrestrial beings is “proved.” Even in the event of confirmation that the signal is of ETI origin, however, the research of university scholars may not prove reassuring to some sectors of the public. Agreement by scientists that the signal does indeed come from another civilization would be a positive outcome, for most – but assurance of the existence of extraterrestrials is not necessarily a positive development, for some.

Historic Examples

A number of space-related events and discoveries, while not closely analogous with an encounter between humanity and an ETI transmission, are nevertheless instructive in showing the extent to which a dramatic discovery enters the educational mainstream. These include the USSR’s launch of the satellite *Sputnik* in 1957 and the discoveries of Pluto, quasars, and pulsars. The former triggered a wave of educational reform in the US (due as much to political as to scientific considerations), whereas the latter have quietly and rapidly made their way into textbooks. Certain returns from space explorations, including photos of the outer planets and satellites by the Voyager space craft



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(and especially whole-planet photos of the Earth by the Apollo astronauts) have become standard icons in classrooms and museums. These have profoundly influenced our views of ourselves and the teaching of science.

Percival Lowell’s conjectures about civilization on Mars shaped the conduct and discussion of planetary science throughout the first few decades of this century. This episode may be exceptionally instructive as it involved widespread public awareness of the possible existence of intelligent beings elsewhere, seemingly documented by scientific observations. The extent to which it influenced textbooks and education is not known to us. It would seem to provide the greatest promise of any recent episode for revealing responses of people and modern institutions to “knowledge” of the existence of ETI’s.

Study of the rates at which appropriate discoveries and conjectures have crept into classrooms and public exhibits and the extent to which they have become accepted as common knowledge via those avenues could provide an excellent indication of the likely effects of an ETI detection on educational institutions and, through them, on the public.

Recommendations on promising actions and avenues for further study are listed at the end of this chapter.

News and Information Industries

American adults obtain most of their day-to-day information on scientific developments from news media such as radio, television, and newspapers. These daily exposures are supplemented by less-widespread reading of science magazines (for example, *Discover*, *Popular Science*, *Scientific American*, and *Sky and Telescope*) and by visits to displays, attendance of lectures and participation in other educational events as noted above. News media provide frequent updates at a (usually) basic level of explanation, with magazine and museum exposures providing less frequent but more detailed updates. The following assesses news-delivery media, with emphasis on journalism.

Journalism and SETI

From the perspective of journalism and media, the disclosure of detection of coherent signals from a civilization on another world would be an “ordinarily sensational event.” As such, the receipt of intelligent signals would be handled journalistically like any other fascinating event in times when startling

announcements are normal. The fact of a possible detection would probably leak out the moment that people in the primary SETI discovery group asked other observatories for verification of a newly-detected signal. Enormous interest would prompt a torrent of questions from those specialized journalists who are prepared for such an event and from a worldwide horde of other journalists with little or no preparation.

The resulting flurry of media attention would resemble that attendant upon any major sports event, or political convention, or major space mission, where the principal actors – athletes, politicians, test pilots, scientists, or engineers – must ration the time they can spend with journalists posing questions on behalf of the public. As a practical consequence, most of the press interactions would be delegated to management figures and other expositors, who would need to be well briefed on many technical matters to avoid creating misunderstandings of fact or emphasis.

Fast and Entertaining

The word “journalism” speaks of time and speed. It is defined as “reporting of events that have happened recently.” It deals with news events that were disclosed recently, often only a few minutes ago. Sometimes the events happen right in front of you, as when the shuttle Challenger exploded or when Neil Armstrong hopped down the ladder onto the lunar desert astronomers call the Sea of Tranquility.

The products of journalism are short, telegraphic, and simplified. They use a few facts and details to stand for many. Journalistic reporting aims at a wide audience rather than specialists. The news is usually addressed to

those who were not present. It is a caricature, with all the brutal simplification – and more than occasional sharpness – that goes with cartoons.

The products of journalism are consumed by people with limited time. The audience takes its news when and where it wishes – in the car, on the train, at the dinner table, on the beach, in the bedroom or living room or bathroom. Journalistic media, including newspapers, magazines, radio, and even television, are highly portable. People use the products of journalism to study role models, to expand their intellectual horizons, to change their world views.

Journalism is a particularly suspect practice as seen from the perspective of scholars, scientists, engineers, and others who deal with complex technical information. Few journalists specialize in reporting science and technology and few of their supervisors reached high rank in the news professions by developing familiarity with discoveries and inventions. The intellectual distance between methodical science and fast-acting journalism has left every scientist or engineer with favorite examples of journalistic incompleteness, distortion, or errors.

Given such experiences it is natural for scholars, scientists, and engineers to worry about extremes of distortion in journalistic reporting of an ETI detection and perhaps even to anticipate the type of reaction that followed Orson Welles' famous radio adaptation of H. G. Wells' story, *The War of the Worlds*. In the instance of a verified ETI detection with all of its sensational implications, distortion of the information with consequent confused or negative public responses would seem to be distinctly possible. This possibility notwithstanding, a 25-year history of interaction between the press and agencies conducting "big sci-

ence" – particularly NASA with its space missions – has served as an admirable dress rehearsal for the Big One, a media response to an announcement of ETI detection. This interval of rehearsal and accommodation is described in the following.

Reporting Science

Scientists and Journalists: Controlled Interactions.

One major route of communication to the public about SETI is through the growing community of science journalists around the world. These journalists work through print and broadcast channels, including magazines, newspapers, radio and television. Many are skilled at extracting background information and news stories from *Spectrum*, *Sky and Telescope*, and comparable magazines, and at employing personal interview practices that ensure reliable communication of accurate information from scientists to the media.

Most major scientific announcements occur under quite controlled and predictable circumstances. They are often keyed to simultaneous publication of a technical article in a scientific journal describing the new discovery. (The media announcement and the scientific publication may also be linked to the issuance of a patent.) Press releases are prepared by publicists for the journals (or the scientists' university or industrial laboratory), usually with the aim of giving journalists a head start of a few days to follow up points in interviews with researchers and others able to comment on the wider significance of the scientific findings. (Occasionally, in hope of a story in prominent publications such as the *New York Times* or the *Wall Street Journal*, laboratory publicists may invite selected journalists to the scien-

tists' laboratory for an extensive advance briefing.) The scientific announcement, when finally made, is often discussed at a press conference attended by reporters from both print and broadcast media.

An important element of this common interaction between scientists and journalists is a "vow of silence" – an agreement by the journalists not to publish the story until the scientific journal publication appears. This is a compromise of journalists' usual rejection on principle of any constraints on the content or timing of their reports. However, there is a payoff, a practical reason for this arrangement; it gives the journalists assurance of the authenticity of the findings to be relayed to the public. Articles appear in scientific journals only after exhaustive peer review by experts in the field. They do not get published if the reviewers have reason to doubt the validity of the new findings. By waiting, journalists can be sure that the validity of the new findings has been confirmed by the people best able to evaluate them.

Some such interaction between press and scientists should certainly be contemplated for a pending announcement that a candidate signal has been confirmed as truly of ETI origin after a prolonged interval of interpretation and analysis.

Direct Observation and "Instant Science" by Journalists

The preceding describes cautious responsible informing of reporters of a momentous science announcement. While this may eventually be required by the SETI research, the project will also be subject to ongoing press scrutiny during less exciting times, while the research is underway. We note as analogies

the many events of technological and scientific significance that have been witnessed directly by reporters. Early examples are the nine Apollo lunar flights. The launches were attended by hundreds of reporters and events of the flight were followed minute by minute via radio communication with the astronauts in space, one of their number in the control room at Houston, and direct broadcast from the moon.

A deeper and more immediate involvement of scientists with reporters began during the explorations of Mars by the Mariner and Viking spacecraft, and of Jupiter, Saturn, Uranus, and Neptune by the Pioneer and Voyager spacecraft during the 1970's and 1980's. A refined mode of interaction and reporting arose at the Jet Propulsion Laboratory. Here the scientific teams working with different instruments aboard the spacecraft reviewed the most significant observations with each other shortly after they were made in preparation for press conferences the next day. This was a new form of rapid "interim publication" of up-to-the-minute findings in the press. The press conferences were supplemented by background briefings at which scientists present for the planetary encounters might well disagree in public about the implications of the findings. Some of the most experienced reporters were given access to the rooms where scientists were receiving data from the spacecraft and were discussing what they were seeing or measuring.

These exposures and the reporting teamwork that evolved as scientists and journalists learned to accommodate each other's needs resulted in a near-perfect reporting of the facts emanating from perhaps our most interesting and exciting planetary mission, the Viking search for life on Mars. The Viking mission and the SETI enterprise have many features in common, as described below.



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Lessons from the Viking Experience

Both Viking and SETI involve fundamental scientific questions in which the public has a sustained deep interest – that is, chemical evolution and the origin and evolution of life. Each project could be labeled a search for life. The kind of heavy media coverage that occurred during the Viking mission can be expected at various stages of the SETI project. As happened with Viking, SETI is likely to involve many instances of trying to explain a body of data that may never provide a definite answer, and whose interpretation involves the impossibility of proving a negative (“no results” does not mean “there are no ETI’s”) to an impatient press. The Viking experience underscores the need for common understanding among a large team of investigators of the numerous pitfalls in explaining such a project in the glare of insistent media attention. The

scientists, engineers, and agencies supporting their work need to anticipate the many pitfalls that can appear and to develop plans for responses that avoid them.

Under the leadership of scientific and project leaders who serve as spokespersons for the enterprise, the technical teams must reach agreement on how they will review their data before release (often very soon after their discussions) and on how they will explain the data in popular terms. As was exemplified by the spokespersons for the Viking findings, project personnel must recognize requirements for openness and non-evasiveness and must avoid expression of opinions where the data are inconclusive. They must be prepared to brief the press at intervals on the scientific background, to recognize the highly variable background of the reporters they meet, and to realize that even their most casual comments in informal settings can become “news.”

The factors of importance during the Viking mission were ably summarized for the Working Group in May, 1992, by Harold P. Klein, the Viking project scientist and spokesperson for announcing each day’s findings and answering questions from the press. In addition to the items in the preceding paragraph, his list of “take home lessons” (which serves as the basis of many of our recommendations) includes the following:

- The project needs a knowledgeable manager or chief scientist to coordinate statements by the project scientists and to be recognized by the media as the project spokesperson;
- The participating scientists must reach an *a priori* agreement regarding dissemination of experimental data;

- The project should hold tutorial sessions for the media beginning before the initiation of the investigation, if possible ;
- Participants must recognize that formal press conferences are not sole sources of information for the media.

Unauthorized Early Disclosures

In reality, another mode of reporting is certain to scoop the controlled release of information as managed by the procedures described above. That is, headlines will quickly appear announcing *Possible ETI Discovery!* every time a SETI observatory calls another to ask for confirmation of a candidate burst of electromagnetic energy coming from some particular location in the sky. Over time, there is likely to be a number of requests for confirmations of these candidate signals (most or all of which will prove to be Earth-based false alarms) and initially, at least, each may trigger a spate of news stories.

The leaks in communication that will trigger these stories are traceable to the SETI verification procedure, which necessarily informs many people that an interesting candidate signal has been detected (see Chapter 1). Some of the persons informed are certain to make the request known in one way or another. One likely dissemination route for the information is via electronic mail. E-mail will be in use by the science teams throughout all phases of the project. There will undoubtedly be a flurry of e-mail messages regarding each candidate signal, with volume increasing if the candidate begins to look like a valid ETI transmission. Each increase in the volume of messages and number of addressees increases the exposure of the information to sources outside the science teams.

The e-mail transmittal of problems with the Hubble Space Telescope provides a rich example of how electronic networks can influence a science mission. As the astronomy community shared early fast-breaking knowledge of the telescope's newly-discovered viewing problems, the information quickly spread in a way that was not possible during earlier missions conducted prior to the widespread use of electronic networks. As a result, the Hubble telescope's problems quickly leaked and became news. SETI investigators can expect to see a similar propagation of information over the networks even if extreme care is taken in the use of e-mail as the verification process occurs. Most verification scenarios will play out to dead ends. However, information passed among e-mail users during normal communications needed to eliminate non-signal false alarms may be unexpectedly publicized with counterproductive effects for the HRMS project. Negative public relations, disenchantment, and loss of credibility are possible outcomes. The dynamics of frequent false alarms and ways of assuring the media and the public that there will be many instances in which candidate signals are discussed and checked for verification or elimination warrant NASA's full attention.

Effects of Media Reporting on Public Responses

Assessing the Response to Sputnik

A salient case of broad response to a major scientific achievement after the Second World War is the worldwide response to the Soviet Union's launch of *Sputnik 1*. The world population was made aware of the event within hours to a few days. Newspapers all over the world headlined the news and editorialized on its meaning for weeks. Public opin-

ion polls had new outlooks to measure for political and scientific purposes: opinions about the competence, depth, intent, and implications of Soviet science and technology.

In the United States, the reactions influenced our law, policy, and educational and governmental institutions, and changed the attitudes of Americans about themselves and their country. The National Aeronautics and Space Act was passed, creating NASA. The National Defense Education Act was passed, creating the first federal funding programs for education. (These concentrated mainly on teacher training and fellowship awards and had little effect on curricula.)

The immediate impact of the news and journalistic coverage of the event was to catapult the existence of Sputnik and its implications into the forefront of issues requiring immediate attention. Europe began to develop a regional concept for cooperation in space ventures. By 1961, a European Launcher Development Organization (ELDO) was formed, followed by a European Space Research organization (ESRO). These two organizations consolidated into the European Space Agency (ESA) in 1975. India, Japan, and China, among other nations, saw the event as documentation of their own relative national weaknesses in science and technology. Within a few years each of them had initiated a national space program that has subsequently expanded to give each a present-day role in space-related activities.

Had we attempted to predict the galvanizing effects of a launch of a Soviet satellite in the 1950's, it is likely that we would have missed on many counts. By analogy, it does not seem likely that media reports of an ETI detection at the present time would ignite a series of responses as profound or as far reaching as these. Partially as a result of the *Sputnik*



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experience, we now know that it is risky to assume that one can foresee all of the implications of a major scientific discovery in advance. The *Sputnik* launch (and other history) teaches us that such events can revolutionize without warning.

The *Sputnik* aftermath was driven by the intense political, military, and economic rivalry that motivated Western democracies and Eastern powers at the time. In the present (1990's) world situation, political, economic and philosophical competitiveness remain strong but the political/military aspects of space-related research as ostensibly innocuous as NASA's HRMS project seem less threatening. Ability to conduct SETI research is not equated in the popular mind to military superiority or proficiency and our assessment of its impact must draw upon other considerations and perspectives. Nevertheless, an announce-

ment in the media of discovery of an extraterrestrial civilization with subsequent intensive reporting is likely to explode across humanity's intellectual and emotional horizons as dramatically as did the announcement of Sputnik.

Do the Facts Matter?

We can be sure that a substantial fraction of the reporting of SETI and any ETI detection will be responsible and accurate. We can also be sure that some of it will be otherwise. We may ask "what difference does it make?"

Does reporting really affect public responses? Does transmission of accurate information to the media result in accurate reporting? In some ways the answer seems to be "no."

Every scientist reads news articles about his or her research with apprehension. Misquotes are expected, as are errors and slanted reporting. Media are motivated by commercial and political pressures as well as by their primary mission of reporting; these considerations put various spins on their presentations of information even when overt errors are avoided. Even if the facts are presented to reporters accurately and without bias, the final practical effects of any strategy to use media as a mode of public information are questionable.

Summaries of the effects of mass communication of scientific work via news media are very few. The most focused major cases involve measurement of knowledge, attitudes, and actual (buying and voting) behavior in human health-related issues. The following examples can be cited:

- (1) Changes in cigarette sales and in smoking behavior after reports of the relationship between cigarette smoking and lung cancer;
- (2) Changes in diet and exercise behavior after reports of evidence of links between cholesterol, fat, and heart disease;
- (3) Public acceptance of polio vaccines, birth control pills, and blood-pressure control medications;
- (4) Polls and referenda in many jurisdictions about nuclear power. (During the referenda, public opinion is sounded repeatedly and the efficacy of campaign messages in changing opinions is tested.)

Evidence of the effects of informal mass reporting and communication has not been compiled and analyzed in ways that enable us to gauge the efficacy of various strategies of formal education. One of the most sustained efforts is that of Jon Miller of Northern Illinois University, who polls national samples at intervals on behalf of the National Science Foundation for a regularly-appearing chapter on public awareness in the series *Science Indicators*. His work is now finding a parallel in Europe under the leadership of John Durant, the first professor of science journalism in Britain. These preliminary efforts may provide a point of entry for NASA.

Journalism and the Social Context

The preceding deals with scientific reporting and its effects on informing the public. There is another important effect of reporting, which is to strongly influence the "emotional context" in which people live and work (as defined in Chapter 3). In an environment of pervasive optimism, the outlooks and actions of individuals and groups are quite different from their responses to identical stimuli encountered in an environment of pessimism. The tone of media reporting has significant

influence on public perceptions of whether times are “good” or “hard.” Conversely, the tone of reporting of the same events is itself likely to vary according to whether the national emotional context is optimistic or pessimistic, usually in reinforcement of the national mood. Given that false claims are almost certain to be reported prematurely as “likely” detections (or as “detections”) and that considerable uncertainty is sure to follow a genuine detection as researchers posit alternative non-ETI hypotheses to account for the signal, the potential for reinforcing a negative emotional context (or undercutting a positive one) seems high.

It is important to recognize that in societies with advanced media technologies and a relatively autonomous and unregulated press, the information media are crucial both in *reflecting* those societies’ emotional contexts and in *establishing* those contexts. For example, by regularly reporting bad news about an economy that is actually in recession, media reflect and highlight real conditions. However, by selecting (and giving prime time and space to) news stories and information about a poor economy, media also exacerbate and maintain general pessimism and lowered consumer confidence.

In the US, television, electronic information systems, and print media pervade the culture. Dissemination as well as constant analysis and interpretation of information and “news” (whether accurate or not) are extraordinarily rapid and widespread and “being informed” is seen as a virtue by a large segment of the public. Thus the media’s selection and deselection of information for dissemination and emphasis, the repetition and privileging of certain stories, the many forms and modes of editorial interpretation, and the quasi-entertainment value of reporting all help to create and maintain the public’s general mood and the

tone of the society in which we live. In these ways, the media are central to establishing and reinforcing a society’s emotional context.

Tabloid and Pseudoscientific Journalism

Certain media productions (both print and audio-visual) imitate the relatively objective forms of journalistic reporting but have no allegiance to accurate content. These media products – tabloid newspapers such as *The National Enquirer*, *The Star*, and *Weekly World News* and television “news magazines” such as *Hard Copy*, *A Current Affair*, and *First Edition* – are consumed by a large segment of the population. Their reportage is profit-driven and entirely motivated by circulation concerns. Their strategy is to exaggerate and sensationalize social or cultural events and concerns, extrapolating and imaginatively embroidering actual events (or completely fabricating them) with an eye to the fears and prurient interests of the public.

From a scientific perspective and in relation to SETI, these exaggerations and inventions are outrageously oblivious to empirical data and actual events. ETI is a particularly popular topic. The tabloid media regularly feature “news stories” that report UFO sightings and direct ETI encounters, often providing the hard photographic evidence of “unretouched” pictures to back up the stories. At the time of this writing, issues of *Weekly World News* highlighted photographs of a space alien descending the steps of a space shuttle, then conversing with presidential candidates. These stories are regarded as sources of real information by some people and as entertainment by others.

Whatever their function or however much the public believes their stories, it is important to

recognize how pervasive tabloid reporting is in US and British society. In the US, the leading weekly print tabloids out-perform most daily newspapers in sales and (presumably) in readership. They generate an ongoing stream of dubious or invented "facts," events, and images that not only form part of the culture's image repertoire (see Chapter 3), but which may also be confused or conflated in many people's minds with the empirical data reported more objectively by responsible news media. It is worth noting that in recent years tabloid reporting has increasingly influenced the mainstream news media. The latter have been forced to investigate and respond to the former's sensational scoops (particularly in relation to tabloid stories with some political content). Recent tabloid stories about the British royal family provide an example.

With respect to the SETI project, we emphasize that there will be no predictable relationship between the actual events of search and signal detection and what the tabloids are likely to report. Although a genuine ETI signal detection might in fact be positively boring compared with the spectacular accounts of ETI detection regularly described by the tabloids, an actual discovery would provide grist for tabloid journalism's sensationalist mill of on-going stories about ETI's for weeks, months or even years after its announcement.

Perhaps the most corrosive effect of tabloid reporting might occur during a long period of non-detection. One likely theme for both the tabloids and television "news magazines" is "*Government Covers Up Big ETI Discovery!*" Conspiracy theories are a stock in trade for tabloid journalism. Tabloids are likely to report that a detection has actually been made but not announced, thereby playing to one of the most frequently-asked questions about SETI: "If they find anything, will the govern-

ment keep it a secret?" Tabloid stories exploiting an interval of non-detection may have potential for undermining public support of SETI research or confidence in government and/or science in general. This possibility (and other considerations) suggests that procedures for periodically announcing and verifying non-detection be as meticulously crafted as procedures for reporting a detection.

Some note must be taken of various extremely popular pseudoscientific accounts of SETI and ETI-related topics. These include books and television programs represented by works like Von Daniken's *Chariots of the Gods* and Streiber's *Communion*. These take a low-profile pseudo-anthropological or unsensationalist view or tone. They are often cited as authoritative texts containing indisputable evidence in support of various pseudoscientific claims by persons attracted to pseudoscience in our culture.

We currently know of no studies of the real impact of tabloid reporting or pseudo-scientific accounts on public perceptions, nor do we have a sense of the nature of their effects upon the critically important emotional context or mood of the nation. Nonetheless it is clear that tabloid journalism and pseudoscientific accounts of phenomena are culturally pervasive and engage a large segment of the population.

While we need to be aware of these media and their influence upon the image repertoires of society, it is probably impossible for NASA or the scientific community to change or influence the method, style, or accuracy of their reporting in any way. Ultimately the only defense against such inaccuracy and sensationalism is the continual dissemination of meticulously accurate and comprehensible information to the public through more responsible channels.

Recommendations based on the considerations discussed in this section are summarized at the end of this chapter.

Entertainment Media

Introduction

The exposure of the public to new information provided by news media and educational institutions plays against a vast influential backdrop provided by entertainment media. Although most people did not watch *Star Wars* to learn about science or extraterrestrial life, they most definitely left the theaters with powerful new images that colored their senses of what is possible, impossible, probable or unknown. Science fiction novels, fantasy artworks, Star Trek conventions and many other enjoyable modes of expression all leave their imprints on human hopes and views of the cosmos. The vast and appealing torrent of information, impressions, and misinformation generated by entertainment industries is a force to be recognized and reckoned with in evaluating likely public responses to an ETI detection. In the following, we assess the entertainment industries for insights on their likely roles in an ETI detection scenario.

The Breadth of Entertainment Media

Entertainment media are audio-visual, print, and broadcast media that are directed toward the public for its aesthetic enjoyment and entertainment. These media may be motivated by economic considerations or may be non-commercial. They may be directed toward a restricted elite, the public in general, adults, or children. Their imagery, sound, poetry, and narrative design may follow the logic of realism, but may also be organized as



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expressionism or fantasy. Entertainment media may also be informative, but this is not their primary function.

Representative entertainment media are shown in Table 5-2. These are not the only venues for expression of speculation about ETI and SETI. Situation comedies or variety shows that draw upon current events, animated series like *The Simpsons*, and courtroom drama series like *LA Law* may all potentially refer to (or dramatize) SETI-related scenarios.

Aesthetic and entertainment media and their productions are characterized by several important features, noted as follows.

- (1) There is no necessary connection between empirical fact (in this instance as defined by the scientific community) and works of

Medium	Examples Related to ETI	Author or Performer
Science Fiction Films and Other Theatrical Films	<i>Close Encounters of the Third Kind, ET</i>	Spielberg
Commercial TV Feature Films	<i>Star Trek: The Next Generation, Alf, V, Intruders</i>	Various
Poems, Short Stories, Novels	<i>Childhood's End</i>	Clarke
"Reality Fictions"	<i>Communion</i>	Streiber
Visual Art	Science Fiction and Fantasy Magazine Covers	Various
Cartoons	<i>Calvin & Hobbes, Robotman, The Far Side</i>	Waterson, Meddick, Larson
Classical Music	<i>Thus Spake Zarathustra</i>	Strauss (in Kubrick's 2001)
Popular Music	<i>Purple People Eater</i>	Wooley
Rap	(no ETI-related Examples Known)	
Live Theater	<i>Visit From a Small Planet, The Search for Intelligent Life in the Universe?</i>	Vidal, Tomlin

Table 5-2. Modes of Artistic Expression Used by Entertainment Media, with Examples Related to SETI

the imagination, nor is there any particular allegiance to accurate portrayal of facts;

(2) The works appeal to an audience's emotions and aesthetic sensibilities, not necessarily to its rational judgment;

(3) In the general realm of mass mediation (somewhat less so in the realm of elite

and/or individual artistic production), the primary goal of presentation is economic profit. This strongly influences the nature and exposure of works made for aesthetic and entertainment purposes.

For reasons of brevity, what follows emphasizes only a few of the artistic and entertain-

ment media listed above, emphasizing those that are of the greatest significance to the SETI enterprise and ways in which they have reflected and shaped the cultural perceptions of their times.

Salient SETI-Related Works and Developments

Imaginative speculation that provided mass entertainment specific to SETI and ETI is traceable to the 1860's. That decade saw the first publication of Jules Verne's popular *Voyages Extraordinaires*, narratives that featured travel to other worlds, contact with different intelligent life forms, and projection of future developments in science, technology, and communications. These were followed in the 1890's by the science fiction works of H. G. Wells. The invention of cinema in the mid-1890's led to several early science fiction films. Most famous among them was one by a French fantasist, Georges Méliès, an adaptation of Verne's *A Trip To The Moon* (1902).

The interval from the 1860's to the early 1900's was a period in which there were radical transformations in scientific knowledge, technology, and communications media in Europe and the US. These greatly affected not only philosophical and artistic thought but also popular conceptions of time, distance, speed, and notions of existential "presence" (Kern, 1983).

The ensuing 1920's – 1930's witnessed the Weimar period in Germany, which was culturally distinguished by the rise of fascism and economic depression. German culture was distinguished at that time by artistic and popular cinematic fascination with horrific and powerful non-human figures with evil designs, including golem, homunculi, vampires and

robots (Eisner, 1969; Kracauer, 1969). In the US the 1930's were generically marked by the emergence of Hugo Gernsback's popular pulp science fiction magazine *Amazing Stories*, featuring gleaming technofantasies as well as tales of alien bug-eyed monsters. The decade culminated in the 1939 World's Fair, which featured technology and futuristic progress as redemptive in relation to national recovery from the Depression (Ross, 1991).

The 1950's were marked by Cold War tension, fear of Communist invasion, and post-A bomb ambivalence about the progressive aspects of high technology in the US. This decade has been called the Golden Age of science fiction cinema. Films about space travel, alien invasions, and ETI encounters were presented in ways that manifested a love-hate dependency upon the military-industrial complex and the scientific-technological elite (Sayre, 1978; Biskind, 1983; Sobchack, 1987; Lucanio, 1987). In Japan, the mid-1950's started the recovery period following the atomic bomb attacks on Hiroshima and Nagasaki. *Godzilla*, *King of the Monsters*, won awards and broke box office records all over Japan in 1954. Continuing through the mid 1980's, science fiction underscored Japan's transformed attitudes toward high technology and was marked by the emergence of the long-term, ever-changing, and sociologically interesting *Godzilla* series and its offshoots.

During the late 1970's and continuing to the present time, science fiction film and literature re-emerged as the popular genres best able to map the American culture's transformed perception of technology, national supremacy, and relationships to other-than-American beings. The works reflected America's heightened mistrust of "rational" governmental and corporate technocracy, the rise of New Age mysticism, a general receptivity to the idea that humans are not the only sentient life

forms in the universe, and feminist and multiculturalist influences on artistic and popular conceptions of "otherness" (Sobchack, 1987).

Thus science fiction works have vividly portrayed humanity's worldly hopes, fears and preoccupations over the decades and have in the process created their own compelling imageries and perspectives on possible ways in which humans and extraterrestrials might interact.

Science Fiction, SETI, and Public Responses

Science fiction literature and films dealing with SETI-related issues have a long tradition in the US. However, only over the last decade has the genre moved from the margins of cultural respectability into the mainstream. Although most works venture into the realm of the fantastic, some of the print literature has tended to be more responsible to science fact. The following works, for example, treat ETI-related themes with reasonable attention to scientific accuracy:

- * *The Black Cloud* (Hoyle, 1957)
- * *The Listeners* (Gunn, 1972)
- * *Contact* (Sagan, 1985)
- * *The Hercules Text* (McDevitt, 1986)
- * *SETI* (Fichtman, 1990)

As a consequence of the long history of science fiction, most cultures now possess a body of speculation on possible encounters between human beings and intelligent extraterrestrial life, much of it imaginatively and aesthetically motivated. (This is discussed in the image repertoires section of Chapter 3.) These imaginative speculations have established a context for (and in many cases a popular belief in) the inevitability of human-ETI contact. This

speculation is always historically related to significant contemporary issues. Further research into the large body of primary novels, short stories, films and secondary critical literature on this relationship should prove extremely helpful in projecting potential scenarios of response to the detection of a signal.

In summary, on-going production of speculative fictions related to SETI and ETI has characterized the major aesthetic and entertainment media, with increased production over the last decade (most notably in the profit-driven film industry). In general, speculative fiction in the print media tends to be more scientifically informed while the visual media tend to be more fanciful. In addition to reflecting cultural conditions, these works have provided humanity with an immense store of images and expectations regarding extraterrestrial life.

Potential Effects of an ETI Detection

It seems natural to suppose that detection of an ETI signal or message would lead to an increase of popular interest in (and speculation about) the nature and motives of the signal originators accompanied by an increase in the production of speculative fictions across entertainment media. However the latter might not necessarily be the case. While detection might result in increased production of speculative works, it is also possible that the shock of the event, or a dearth or plenitude of information about the civilization of the signal originators, or political controversy surrounding both the signal and potential replies to it, would give artists pause or cause profit-driven mass media to hesitate.

It is worth noting that production of science fiction films in general (and those about space

travel in particular) was extremely low in the decade following the televised moon landings. While there are a number of cultural reasons for this decline, it is also possible that popular interest was redirected from science fiction to science fact regarding space travel and other worlds and that fantastic speculations in the immediate face of new information could not sustain audience "suspension of disbelief" in films that departed grossly from the newly-perceived reality. After the new information was taken for granted, became less newsworthy, and merged with the general cultural background, imaginative fancy took flight again.

It is also worth noting that the conservative profit-driven film and television industries generally avoid highly controversial material. For example, most films released about American participation in the Vietnam war came about a decade after the controversy began to dwindle. AIDS has only recently been the subject of entertainment media although it has been a major public issue for years. The national battle over reproductive rights is still considered too controversial a subject for the entertainment media (indeed, several commercial sponsors have withdrawn from advertising commitments when television has aired docu-dramas on the subject). In sum, a great deal will depend upon how information-rich an ETI signal is, how much controversy is generated by its receipt, and consequent proposals for human responses. It is likely that the less information-rich the signal and the less controversial its reception, the more imaginative speculation there will be about its originators and significance across the entertainment media.

SETI research is among a handful of scientific endeavors that have immense inherent appeal as a topic of entertainment. It is likely that representations of SETI in entertainment

media can be made more accurate while retaining popular appeal by closer interaction between the SETI and entertainment communities. HRMS scientists might, for example, convene with major directors associated with science fiction in a symposium to discuss what SETI and an actual ETI detection could mean to the genre. (These directors might include Robert Wise, Ridley Scott, George Lucas, and Stephen Spielberg.) This convention could also assess responses by the public likely to result from media treatments. Other possibilities are outlined in the recommendations.

Recommendations

Educational Resources and Institutions

NASA might be well advised to take the following actions: first as a way of better informing educational institutions that inform the public about SETI; and second, to establish a base of credible information for use in the event of a detection:

- (1) Identify a way in which information that would be useful to school teachers for presenting and displaying SETI information in their classrooms can be made available to them. (This could include identification or even publication of an appropriate book, which would also be recommended to librarians; see 5 below);
- (2) Make updated lists of reliable SETI books and materials available to librarians (possibly through their professional associations and networks);
- (3) Establish a formal line of communication from the SETI community to leading pro-

fessional organizations of teachers (for example, the NSTA) to be used for routine updates about SETI and in the event of a detection.

- (4) Identify ways of informing the public that scientific disagreement and debate over the nature of a candidate signal is a normal and expected feature of scientific dialog;
- (5) Survey elementary school teachers to learn what information about SETI (and the forms in which it should be made available) would be most useful to them;
- (6) Study the ways in which the dramatic examples noted – Sputnik, Pluto, Lowell’s views of Mars – influenced textbooks and educational institutions.

News Media and Related Media

The following are ways of improving public understanding of SETI as modulated by the media.

- (1) Find a timely and routine way of keeping the public and media informed about any candidate signals checked and discarded or checked and still under scrutiny;
- (2) HRMS news releases should regularly remind the public that dispute and debate among scientists is a normal feature of science and that such should be expected if an ETI detection is announced (this parallels a need to convey the same message through educational processes; see 4 above.);
- (3) Devise a formal mechanism for release of information to the media, drawing heavily upon the Viking experience;

- (4) Conduct studies of historic science stories that have resemblance to an ETI announcement for insights on which communications were effective, which were not, and how the public responded. These might include...

- * The Viking search for life on Mars;

- * The Apollo 11 landing on the Moon;

- * The discovery of pulsars (whose regular radio pulses at first were considered possible signals from extraterrestrial civilizations);

- * The eclipse in 1919 that tested and confirmed the warping of space-time in Einstein’s theory of relativity;

- * Reports of Lowell’s assertions that intelligent beings had established a civilization on Mars;

- * The passage of the Earth through the tail of Halley’s comet in 1910;

- * A detection by Soviet astronomers of a secret military satellite and their mistaken description of the signal as an ETI beacon in ensuing publications.

(The effects of reporting these stories parallel their effects on educational institutions; see 6 above.)

- (5) Identify and understand the responses of peoples of different cultures to reports on the Apollo and Viking missions, in particular their reactions to news of the Apollo and Viking landings. Were there differences between cultures in the ways that the activities were presented? In the points emphasized? How did people respond to different modes of presentation, to the activities themselves, and to the scientific

discoveries? (This recommendation supports a need for information of the kind identified in Chapter 3.)

- (6) Devise a way of regularly reporting the absence of detected signals for the part of the search that precedes a genuine detection and to report other (unanticipated) astronomical discoveries resulting from the project;
- (7) Designate a standby team of scientists, journalists, and others who will respond to exaggerated or erroneous reports where the latter appear to be confusing the public (see Recommendation 5, Chapter 3);
- (8) Devise a method of regularly reminding the public that false alarms (detections of transient radio noise or signals from human sources) will be a frequent and ongoing result of SETI research.
- (4) Briefing materials made available to news and information media should also be made available to entertainment media people.
- (5) HRMS/NASA technical assistance might be made available during film and television productions of relevance to SETI.
- (6) NASA should regularly invite science fiction writers to briefings for the press and VIP's, as JPL started to do with the later Voyager flybys.
- (7) NASA should invite representatives from major entertainment media to serve on committees that plan the HRMS's public relations/public information efforts.
- (8) A summary should be compiled of scenarios to ETI detection as envisioned by works of science fiction.

Entertainment Media

- (1) HRMS staff should establish liaison with the entertainment industry, both to educate key media artists and producers and for HRMS staff to learn about communicating through popular forms. (For example, a video game based upon SETI could be exciting, informative, and accurate.)
- (2) Consider encouraging a filmmaker to develop a major feature about SETI and receipt of an ETI signal with responsible but dramatic treatment.
- (3) Consider offering a non-technical seminar for producers, screenwriters, television executives and programmers to demonstrate how SETI works and what actual detection scenarios might be like.